Advanced Machine Learning

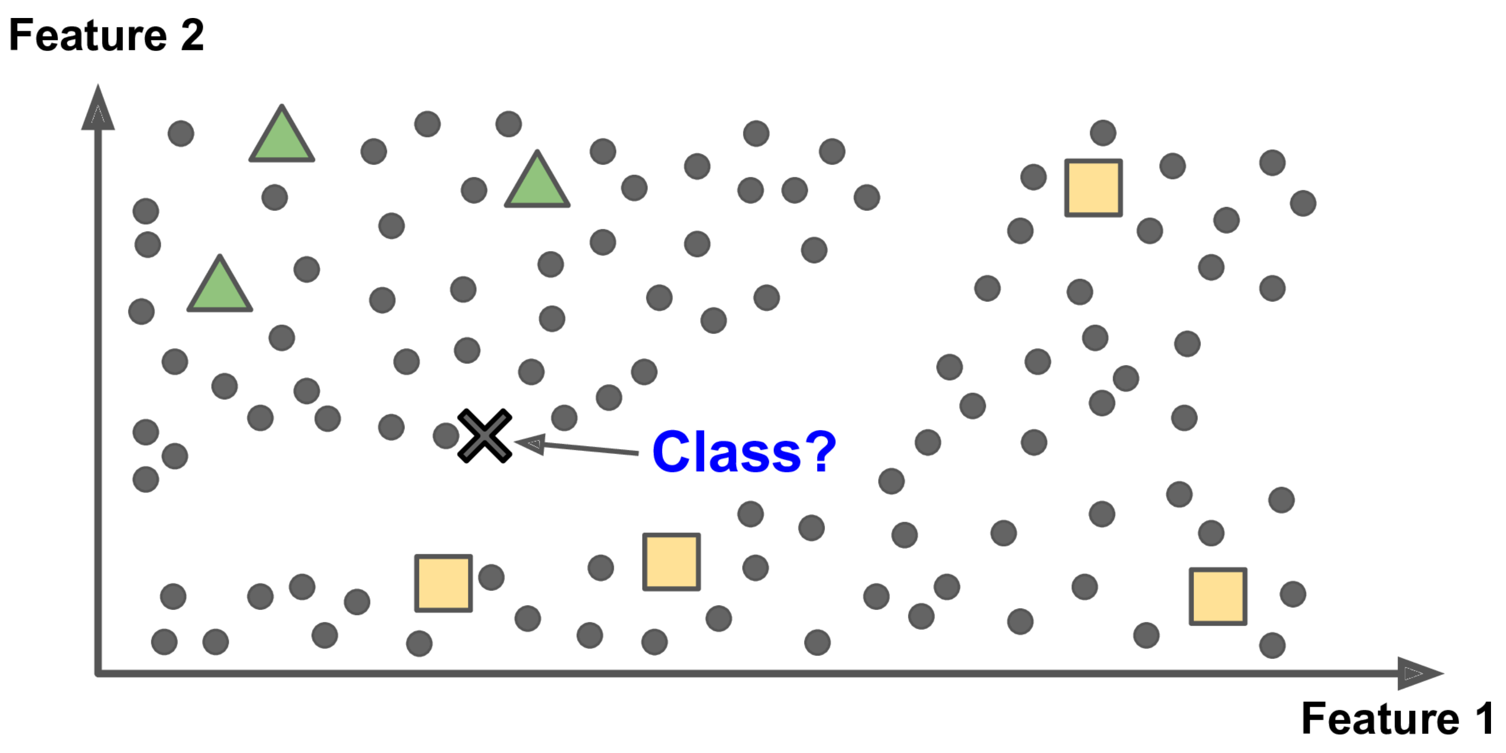
DATA 642

Lecture 10 — Clustering for Semi Supervised Learning

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Central Idea of Clustering for Semi-Supervised Learning

Leverage the structure within unlabeled data to enhance supervised learning in scenarios with limited labeled data.



- **Methodology**:
    - Cluster unlabeled data points based on similarity or proximity in feature space.
    - Propagate labels from a small set of labeled instances to entire clusters.
    - Augment the available labeled data by assigning labels to all instances within the same cluster.
- **Benefits**:
    - Utilizes the abundance of unlabeled data to improve model performance.
    - Enhances the robustness and accuracy of supervised learning algorithms.
    - Reduces the reliance on large labeled datasets, which can be costly or difficult to obtain.
- **Applications**:
    - Text classification
    - Image classification
    - Anomaly detection
    - Customer segmentation
    - Various other domains with limited labeled data availability.

Using K-Means for Semi-Supervised Learning

1. **Cluster Unlabeled Data**:
    - Initially, unlabeled data points are clustered using the K-Means algorithm to identify clusters of similar instances. Each cluster is represented by its centroid.
2. **Assign Cluster Labels**:
    - Labels are assigned to each cluster centroid based on majority voting or using labels from a small set of labeled instances that are closest to each centroid.
        - Majority voting: The most frequent class label among the labeled instances within a cluster is assigned to the cluster centroid.
        - Closest labeled instances: The labels of the nearest labeled instances to each cluster centroid are assigned to the centroid.

3. **Propagate Cluster Labels**:

    - Once cluster centroids are labeled, the labels are propagated to all data points within the same cluster.
    - Each data point in the cluster is assigned the label of the corresponding cluster centroid.

4. **Augment Labeled Data**:

    - After label propagation, the unlabeled data is effectively labeled based on the clustering results.
    - The augmented labeled dataset now contains both the original labeled instances and the propagated labels from the clusters.

5. **Train Supervised Learning Model**:

    - The augmented labeled dataset is used to train a supervised learning model (e.g., classification or regression).
    - The model learns from both the original labeled data and the augmented data obtained through label propagation.

6. **Evaluate Model Performance**:

    - The trained model's performance is evaluated using standard evaluation metrics on a separate validation or test dataset.
    - The model's ability to generalize to unseen data is assessed to determine the effectiveness of using label propagation for semi-supervised learning.